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Course: ECE 601
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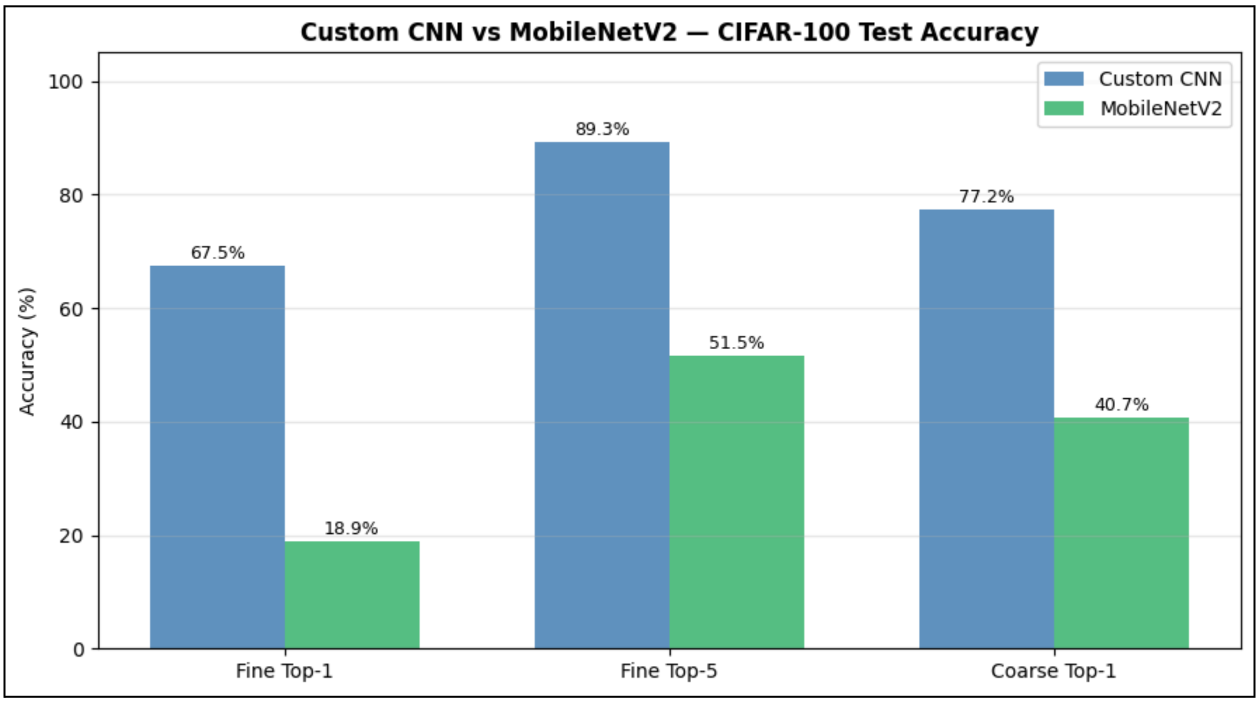
Project Phase 5

Phase Introduction:

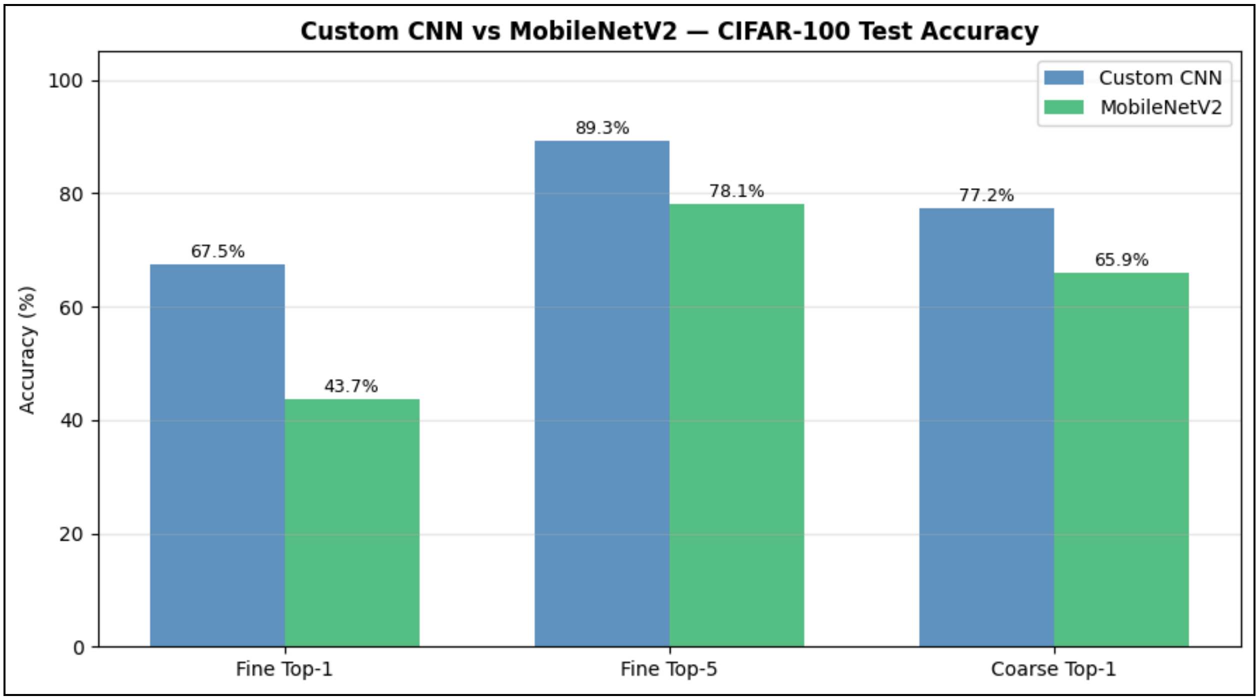
In this phase MobileNet V2 was implemented for training and evaluation on the Cifar-100 Fine and Coarse classification tasks. This was done to create a numerical comparison between a baseline lightweight CNN and the custom CNN I built in this project.

Numerical Comparison:

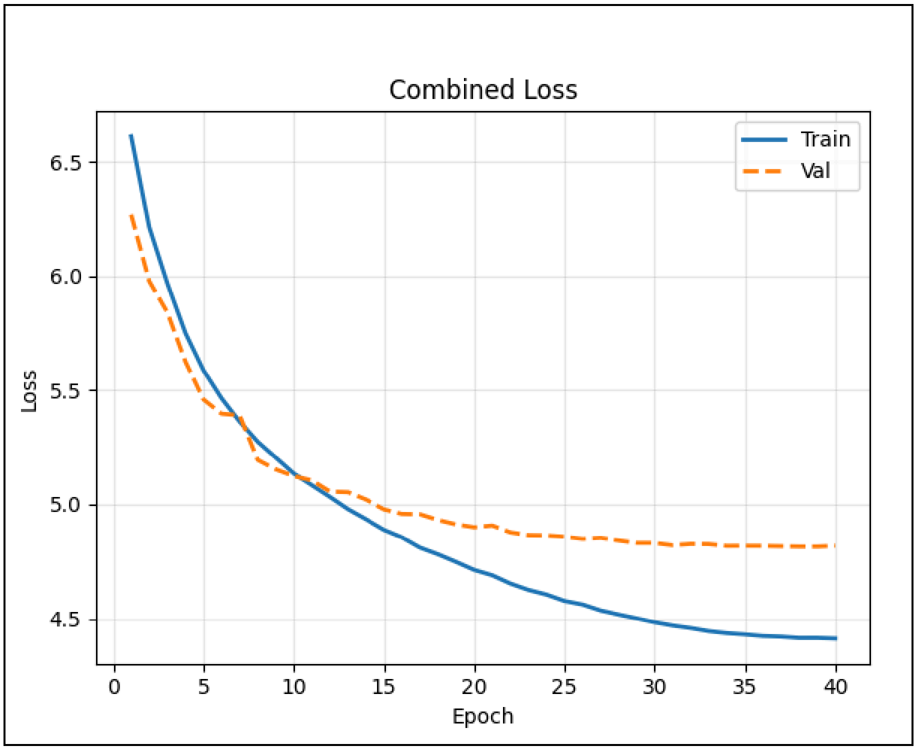
Custom CNN Parameter Count: 2,748,600
MobileNet V2 Parameter Count: 3,265,656



These performance statistics show that the MobileNet implemented with ~500,000 more parameters is much worse for this set than the Custom CNN. The MobileNet performed 48.56% worse on the Fine (100 class) classification task. It also performed 36.58% worse on the Coarse (20 class) classification task. These results were from the initial run where the MobileNet was only trained for 20 epochs which was half as much training as the Custom CNN received.



These are the results when I doubled the amount of training epochs from 20 to 40 for the MobileNet so that both the MobileNet and Custom CNN were trained for an equivalent number of epochs. Here the MobileNet made significant progress purely through extended training time, here the MobileNet only performed 12.3% worse on the Coarse task and 23.8% worse on the Fine task than the Custom CNN. By doubling the training time the MobileNet performed over double (from 18% to 43%) as well on the Fine task. This level of improvement indicates that a longer training time would push the MobileNet even closer to the performance of the Custom CNN. For reference this is the training curve for the Custom CNN which indicates it may have plateaued in terms of performance;



Qualitative Comparison:

The MobileNet V2 was loaded with the trained weights from ImageNet1k which had;

- 71.8% Top-1 accuracy on the ImageNet1k set, that is 71.8% on a 1000 class classification task.
- 90.2% Top-5 accuracy on the ImageNet1k set.

The level of fine tuning that was done for strong performance on the ImageNet dataset did not transfer well at all to the Cifar-100 dataset. This can be attributed to a data mismatch which has to do with the resolutions of the different datasets. ImageNet images are 224x224 while Cifar-100 images are 32x32. This means that the filters formed for specific feature extraction from the ImageNet dataset were likely blind to the same features in the Cifar dataset. What I mean is that a feature in the ImageNet dataset that is represented by 10 pixels may be represented by 1 in the Cifar dataset which is why the performance strength did not translate well between datasets.

It is notable to notice that the gap in numerical performances between the Custom CNN and the MobileNet in the Fine task is almost double that of the Coarse label. This also indicates that the resolution is an issue for MobileNet as it performs much stronger on the more general task. This is because MobileNets have multiple layers towards the beginning which downsample the images, meaning the 32x32 image resolution decreases very quickly. The Custom CNN does not have this issue because it does not reduce spatial dimension anyway other than pooling layers.

MobileNet V2 has more than 50 layers compared to the custom CNN which has less than 10 layers. Due to this, the 3.2 million parameters of the MobileNet are spread out across a lot more layers than the 2.7 million parameters of the custom CNN. This is due to the depthwise separable convolutions which perform convolutions with less parameters than the traditional convolutional layers, however, the tradeoff of increased depth is that the depthwise separable convolutions have less modeling strength than traditional convolutional layers. This is possibly reflected in the models poor performance on the Fine task. It is possible that the loss in modeling strength cannot be compensated for when working with a resolution so small, specifically when the initial striding and downsampling is intended for a much larger resolution.

References:

[1] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4510-4520. [Online]. Available: https://arxiv.org/abs/1801.04381